Gerardo Alberto Amaya Fuentes

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LinkedIn | GitHub | Portfolio

Professional Summary

Full Stack Developer with 4 years of experience building digital government platforms in El Salvador, including online procedure portals for the Ministry of Defense, the Fire Department and municipal governments, used by citizens and businesses. Specialized in backend development with NestJS, Laravel and PostgreSQL on Google Cloud, with experience in 3-D Secure payments, electronic document signing, AI agents and OWASP-based security testing. I have built AI applications with Python, FastAPI and the Claude API (Anthropic), such as a conversational trip planner and a system that turns scanned invoices into accounting data using OCR and LLMs.

Professional Experience

Government of El Salvador | San Salvador, El Salvador

Full Stack Developer | Apr 2023 - Present

- **Ministry of Defense (MDN) procedures portal:** portal to apply for and track permits for hazardous materials, such as explosives, online. Backend built on NestJS microservices and PostgreSQL, with JWT/JWKS authentication, per-case access control, online payments, scheduled jobs with Cloud Scheduler to notify applicants, and monitoring with Cloud Logging.
- **Fire Department (CBES) procedures portal:** portal for businesses to process fire prevention inspection certifications online, including inspections of vehicles that transport hazardous materials. NestJS backend with a procedures gateway and per-institution adapters, dynamic appointment scheduling for inspectors, document review with observations, deadlines and automatic notifications, PDF resolutions with electronic signatures, and lookups against RNPN (national ID) and the Ministry of Finance (tax ID). Includes a Gemini-based AI agent that applies the document rules, classifies applicants and generates responses to users.

- **Distrito Digital La Paz Este:** Laravel platform on Google App Engine that digitizes six municipal procedures (solvency certificates, signage, street opening, minor works, major works and antennas) across three municipalities, so citizens can complete them without going in person. Citizen portal and role-based internal portal, national ID validation, online payment through Serfinsa or in-person payment, resolutions signed with Uanataca and archived in Cloud Storage, configurable fees per district, and scheduled tasks with Cloud Scheduler.
- **School Escalation System (SECE):** platform to survey and track around 1,400 schools ahead of the delivery of school supply packages. Monitors assigned by route record classrooms, sections, shifts, teachers, infrastructure, connectivity and incidents on site, and a dashboard shows, by department, the schools that need attention.
- **Electronic signature:** PBS signing system and integration of the Uanataca One-Shot API to digitally sign documents; the Optimizer is self-hosted in Docker so documents and personal data never leave the institutional network.
- **Serfinsa payment gateway:** centralized payment broker designed to be consumed by multiple procedure portals. 3-D Secure authentication with Cardinal (merchant JWT), asynchronously queued charges with status lookup by identifier, attempt logging in Cloud SQL (MySQL), cardholder validation against RNPN, and security headers (HSTS, X-Frame-Options, X-Content-Type-Options); payment flow failure diagnosis with Cloud Logging.
- **Security and pentesting:** automated Jest suites based on the OWASP Top 10 and OWASP API Security Top 10 covering BOLA/IDOR, BFLA (role-based privilege escalation), SQL and NoSQL injection, XSS, broken authentication and JWT tampering, mass assignment, brute force and rate limiting, SSRF, CORS and security headers; resolution of JWT/JWKS authentication incidents in production.
- **Chivo Pets:** landing page for the Chivo Pets Veterinary Hospital, the first public veterinary hospital in El Salvador and a benchmark in Latin America. Responsive site presenting the hospital's services (consultations, 24/7 emergencies, surgery, clinical laboratory and hospitalization).

Asociación Conexión | San Salvador, El Salvador

Full Stack Developer Intern | Jan 2023 - Apr 2023

- Training and certification internship in Java, PHP, Laravel, NestJS and databases.

Personal Projects

Smart Report AI | Python, FastAPI, PostgreSQL + PostGIS, Claude AI, Next.js, Telegram Bot API | 2026

- **Problem:** reporting a public road hazard (for example, a pothole or an open manhole) is slow, and duplicate reports overwhelm the people who handle them.
- **Solution:** citizens report through a Telegram bot in under a minute. A FastAPI service stores each report with its location in PostgreSQL + PostGIS and uses Claude AI to interpret it.
- Reports of the same incident are grouped automatically, and the dispatcher sees them in a Next.js web dashboard as a single case with all the evidence combined.

DocuFlow AI | Python, FastAPI, PostgreSQL + pgvector, Celery, Claude AI, Next.js | 2026

- **Problem:** manually entering paper invoices into the accounting system is slow and error-prone.
- **Solution:** a pipeline that runs OCR on the scanned invoice and uses Claude AI to extract the accounting fields. Each field carries a measured confidence score, so human review focuses on the fields that need it.
- Processing runs in the background with Celery queues, and semantic search over pgvector answers questions while citing the source document.

Waypoint AI | Python, FastAPI, PostgreSQL + PostGIS, Claude AI, Next.js | 2026

- Natural language trip planner for El Salvador, for example: "beach and traditional food, leaving from San Salvador, two hours of driving at most".
- Claude AI turns the request into structured constraints, and a deterministic engine builds the plan from real places and distances in PostGIS, so the AI cannot invent destinations or routes.

Real Estate Lead Management | Angular, TypeScript, Node.js, Express, MongoDB, Playwright | 2026

- Module to register and follow up on a real estate agency's leads, with an Angular frontend and a REST API in Node.js and Express.
- Metrics dashboard computed in a single MongoDB aggregation query, and end-to-end tests with Playwright.

Travel Quote | Node.js, Express, Sequelize, MySQL, Docker | 2025

- System to quote tourist transportation services, with a REST backend in a layered clean architecture, MySQL persistence through the Sequelize ORM, and containerized deployment with Docker.

Education

Universidad Francisco Gavidia | San Salvador, El Salvador

B.S. in Computer Science Engineering, all coursework completed (graduate)

Instituto Técnico Ricaldone | Technical High School Diploma | 2017 - 2019

Certifications

- Associate Cloud Engineer, Google Cloud. Sep 2026, valid until Sep 2029.
- MTA: Database Fundamentals, Microsoft. Jan 2022.
- Scrum Fundamentals Certified, VMEdu. Credential ID 982714.
- Mobile App Development, Google. Jul 2023. Android app development with Kotlin.
- PL/SQL Table Functions, Oracle.
- Google Cloud Skill Badges (8): Cloud Run, CI/CD, IAM, Vision API, load balancing, VPC and development environments.

Additional Training

- SQL Server Specialization, FORMAS BID
- Oracle AWS (basic, intermediate and advanced), INSAFORP
- Backend Developer Bootcamp, INSAFORP and Asociación Conexión
- Advanced Web Programming, INSAFORP and Asociación Conexión
- Java Fundamentals, INSAFORP
- Object-Oriented Programming with Python, INSAFORP
- Data Structures and Functions in Python, INSAFORP
- Python Programming, Presidential Secretariat for Innovation

Technical Skills

- **Backend:** Node.js, NestJS (TypeORM, Prisma), Express, Python (FastAPI, Django, Flask, Celery), PHP (Laravel), Java, C# .NET

- **Frontend:** Next.js, React, Angular, Vue.js, TypeScript, Tailwind CSS
- **Mobile:** Kotlin (Android)
- **Databases:** PostgreSQL (PostGIS, pgvector), MySQL, SQL Server (T-SQL), Oracle (PL/SQL), MongoDB
- **Cloud and DevOps:** Google Cloud (GKE, App Engine, Cloud Run, Cloud Scheduler, Cloud SQL, Cloud Storage, Pub/Sub, Secret Manager, IAM, Cloud Logging), Docker, CI/CD
- **Security and QA:** web and API pentesting (OWASP Top 10 and API Security Top 10), 3-D Secure, JWT/JWKS, Jest, Playwright, Swagger
- **Artificial Intelligence:** Claude API (Anthropic), Gemini, OpenAI-compatible APIs, AI agents, OCR and LLM-based extraction, semantic search
- **Languages:** Spanish (native), English (intermediate)

References

- Ivan Orlando Moran Interiano, Government of El Salvador. +503 6028-8419, ivan.moran@goes.gob.sv
- Gilberto Lara, Asociación Conexión. +503 2249-2307, galara@conexion.sv